

ERIC M. TANG, PH.D.

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SUMMARY

Imaging research scientist with eight years of experience building medical imaging systems end to end — optical hardware, real-time acquisition software, and deep learning. Ph.D. in Biomedical Engineering; first-author publications in Biomedical Optics Express.

TECHNICAL SKILLS

Programming: Python, C/C++ (multithreaded, real-time), MATLAB; PyTorch, TensorFlow, OpenCV

Machine Learning: CNNs (U-Net, YOLO) and GANs for detection, segmentation, tracking, denoising, and super-resolution; dataset curation and annotation QC; model training, evaluation, and optimization

Edge Deployment: PyTorch model export via ONNX, TensorRT optimization for real-time inference on embedded hardware (NVIDIA Jetson Orin Nano); low-latency C++ runtimes (LibTorch)

Computer Vision & Imaging: Camera calibration, stereo vision, 3D reconstruction; signal and image processing; multimodal fusion; OCT, photoacoustic, and visible-light systems

Hardware: Camera and sensor integration, DAQ/FPGA programming, control systems, optical system design

EXPERIENCE

Topcon Healthcare — San Jose, CA

May 2023 – Present

Senior Research Scientist

- Manage and optimize real-time image acquisition software (C++) for device control and high-throughput data capture at 2 gigasamples/second via multithreaded programming
- Designed a physics-informed deep learning model (PyTorch) for super-resolution and denoising, integrating the optical system's PSF; achieved a 2× improvement in resolution and signal quality, presented to executives
- Deployed deep learning models to embedded edge hardware via ONNX export and TensorRT optimization for real-time inference on an NVIDIA Jetson Orin Nano
- Led cross-organizational vendor collaborations to co-develop a novel compact imaging sensor and scale it to production at a 2× reduction in device size

Diagnostic Imaging & Image-Guided Interventions Lab — Nashville, TN

May 2022 – May 2023

Postdoctoral Research Fellow

- Extended a deep-learning pipeline for 4D surgical-instrument tracking, reaching 99% sensitivity at 23 fps via multi-channel inputs and data augmentation
- Supported development and real-time deployment of a multi-scale U-Net denoising framework, improving image quality by 90% at video rate

Vanderbilt University — Nashville, TN

May 2018 – May 2022

Graduate Research Assistant

- Optimized a high-speed multimodal OCT imaging system and managed its real-time acquisition codebase (C++, 4 GS/s) with GPU-accelerated CNN inference
- Integrated automated surgical-instrument tracking using a lightweight detector (YOLOv4) at 120 fps across multi-view inputs

EDUCATION

Vanderbilt University — Ph.D., Biomedical Engineering

2022

Duke University — B.S., Biomedical Engineering

2018

Duke University — B.A., Computer Science

2018

Publications: First-author work in Biomedical Optics Express — full list at [Google Scholar](#).